Determination of Drought Triggering Probability via Bayesian Network

Robert R. Tucci

ICPAC, www.ar-tiste.xyz

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Links

Repo at
`https://github.com/nishadhka/bn-ibf-e4drr`
Includes

- python code (well documented with plenty of docstrings)
- folder entitled `jupyter_notebooks` with 3 notebooks
- folder entitled white-paper with latex code and PDF for **preliminary** white-paper
  `https://github.com/nishadhka/bn-ibf-e4drr/blob/main/white-paper/anticipatory-action.pdf`

I would appreciate it if you would take a look at the code and read the white paper. Comments from why did you do this instead of that, or I think this part is wrong, or BS or too technical or unnecessary or obscure are welcomed and I will attempt to address them.

Brief review of Tim Palmer ECMWF Theory

$I^* =$ impact threshold for drought, it equals minus an SPI (Standardized Precipitation Index)

$\mu =$ ensemble member index

$M =$ number of ensemble members, 51 for *ECMWF*

$\mathbb{1}() =$ indicator function. $\mathbb{1}(a > b)$ equals 1 if $a > b$ and is zero otherwise

$T =$ trigger for drought

$P^* =$ threshold probability that triggers a drought warning

$$\underbrace{P(\underline{I} \geq I^*)}_{\text{probability that impact is greater than threshold } I^*} = \frac{1}{M}\sum_{\mu=1}^{M} \mathbb{1}(I_\mu \geq I^*)$$

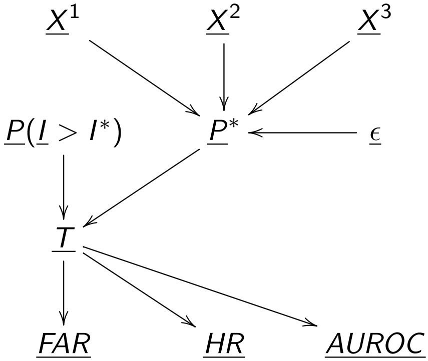
$$T = \mathbb{1}\left[P(\underline{I} \geq I^*) > P^*\right]$$

Problem to solve

Find an automatic method of determining triggering probability P^*.

- The new method should be better than Nishadh's way of doing this for Karamoja, based on FAR/HR/AUROC.
- The new method should take into account **antecedents** like population density, historical catastrophes, soil moisture, etc.

- **bnet (Bayesian network)** (see my free, open source book `https://github.com/rrtucci/Bayesuvius/raw/master/main.pdf`)
- Do bnet calculations using **Pyagrum** software at `https://pyagrum.readthedocs.io/en/` combined with my auxiliary software



$\underline{X}^1, \underline{X}^2, \underline{X}^3$ are antecedents
$\underline{\epsilon}$ is noise

Much Still to Do

- So far, I've only used synthetic data to train bnet. **Need real data to train it.**
- Need to do a lot of work **fine tuning** and **integrating** this software with full corpus of DAA (Drought Anticipatory Action) software
- Need to do **modelling of antecedent probabilities**.